

The Rise and Fall of Behaviorism: The Narrative and the Numbers

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The history of 20th-century American psychology is often depicted as a history of the rise and fall of behaviorism. Although historians disagree about the theoretical and social factors that have contributed to the development of experimental psychology, there is widespread consensus about the growing and (later) declining influence of behaviorism between approximately 1920 and 1970. Because such wide-scope claims about the development of American psychology are typically based on small and unrepresentative samples of historical data, however, the question arises to what extent the received view is justified. This article aims to answer this question in two ways. First, we use advanced scientometric tools (e.g., bibliometric mapping, cocitation analysis, and term co-occurrence analysis) to quantitatively analyze the metadata of 119,278 articles published in American journals between 1920 and 1970. We reconstruct the development and structure of American psychology using cocitation and co-occurrence networks and argue that the standard story needs reappraising. Second, we argue that the question whether behaviorism was the “dominant” school of American psychology is historically misleading to begin with. Using the results of our bibliometric analyses, we argue that questions about the development of American psychology deserve more fine-grained answers.

Keywords: behaviorism, American psychology, bibliometric mapping, cocitation analysis, co-occurrence analysis

The history of 20th-century American psychology is often depicted as a history of the rise and fall of behaviorism, the view that psychology should become “a purely objective experimental branch of natural science” (Watson, 1913, p. 248). Although early 20th-century psychologists aimed to redefine their discipline as a science of behavior, the popularity of behaviorism declined from the late 1950s onward, when psychologists, linguists, and computer scientists joined forces and developed empirical approaches to the study of mind and cognition.

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Although historians of science disagree about the theoretical and social factors that have contributed to the development of experimental psychology (Cohen-Cole, 2014; Greenwood, 2009; Leahey, 2001), there is a widespread consensus about the growing and declining influence of behaviorism in 20th-century American psychology (DeGrandpré & Buskist, 2000; O'Donohue & Kitchener, 1999; Staddon, 1999; Zuriff, 1985).

From a methodological perspective, these claims are contentious, because wide-scope claims about the development of American psychology are rarely backed up by representative empirical data. Although nobody will deny that many of today's best-known behaviorists produced their most influential work between 1920 and 1960, it is unclear what proportion of American psychologists embraced a behavioristic conception of psychology both during and after the heyday of behaviorism. Similar worries can be raised about the historians' conclusions about developments internal to behaviorism. Although most scholars identify Watson, Hull, Tolman, and Skinner as the most influential (neo-)behaviorists,¹ these claims are seldom supported by balanced empirical evidence. In general, it is unclear to what extent the list of psychologists who have been most extensively studied by historians of behaviorism forms a faithful representation of the scholars that actually influenced the development of behaviorism in the first half of the 20th century.

The problem that wide-scope claims about the history of American psychology are rarely backed up by balanced empirical data is especially pressing because it seems likely that historians overestimate the influence of canonized schools and scholars (whether it is behaviorism in the 1930s or cognitive psychology in the 1970s). Not only do historians tend to ignore psychologists who did not belong to any school (Green, Feinerer, & Burman, 2013); in studying the histories of influential schools and scholars, historians also tend to limit themselves to the writings "of a few known authors that are transmitted from generation to generation [...] as *the* authors to read" (Betti, van den Berg, Oortwijn, & Treijtel, 2019, p. 296). More particularly, historical claims about the development of American psychology are often based on corpora that are (a) not clearly specified (which texts exactly did the historian study in arriving at their conclusions?), (b) very small (how many of the tens of thousands of texts produced by psychologists did the historian actually read?), and (c) not representative (how can the historian guarantee that the texts he or she studied form a representative sample of the total body of texts produced by psychologists?). Although in-depth text analyses and archive studies can provide historians with a first indication of the influence of a psychologist, a method, or a school of psychology, their conclusions will always be shaped by the texts and archives they choose to study.

In this article, we aim to transcend the received view about the development of 20th-century American psychology in two ways. First, we use advanced text analysis tools (e.g., bibliometric mapping, cocitation analysis, and co-occurrence analysis) to quantitatively analyze the metadata of 119,278 articles published in American psychology journals between 1920 and 1970. We analyze both the citations and the title terms of these articles and generate cocitation and co-occurrence networks for every consecutive decade between 1920 and 1970 to reconstruct the structure and development of mid-20th-century American psychology and the structure and development of behaviorism. Second, we argue that the question whether behaviorism was the "dominant" school of American psychology is historically misleading to begin with. Using the results of our bibliometric analyses, we argue that questions about the development of American psychology deserve more fine-grained answers.

This article is organized as follows. After a brief outline of what might be called the received view about the history of behaviorism, we explain the main methodological challenges

¹ See, for example, Boakes (1984, p. 237), Smith (1986, pp. 21–22) and Mills (2012, pp. 104–108). Most historians paint a picture of the development of behaviorism in which first Watson and later Hull and Tolman offered the "dominant orientation in American departments of psychology until after the end of the Second War, when it was displaced by [Skinner's] radical behaviorism" (Greenwood, 2009, p. 477).

surrounding quantitative representations of the development of American psychology. Next, we describe our data and methods and provide an overview of our findings about the structure and development of American psychology between 1920 and 1970. Finally, we discuss the implications of our findings and argue that the standard story about the development of American psychology needs reappraising.

Behaviorism: The Narrative

Behaviorism is a complex of methodological, epistemological, and sometimes ontological assumptions about the foundations of psychology. Where psychology is traditionally defined as the study of mental phenomena, behaviorists typically argue that psychology should become a science of behavior. More specifically, most behaviorists agree that (a) psychology is or should become a branch of natural science, (b) psychologists should study behavior instead of mental phenomena, and (c) a science of behavior should be built exclusively on publicly available evidence (thereby dismissing the use of introspection in psychological research). Usually, behaviorists combine this view about the nature of psychology with a set of empirical assumptions—for instance, the assumption that the behavior of an organism is determined by the organism’s reinforcement history. Outside these shared philosophical and empirical commitments, behaviorists also strongly disagree about a wide range of issues. Most importantly, they disagree about the domain for psychology (should a study of behavior include or exclude physiological variables?), the nature of the observation language (should behaviorists tolerate intensional descriptions or purposive language?), and the types of theoretical concepts allowed in the construction of behaviorist theories (are intervening variables or hypothetical constructs acceptable?).

Historians often distinguish between two types of behaviorism in psychology: methodological and radical behaviorism (Day, 1983; Mills, 2012; Moore, 1981).² Methodological behaviorists view (a)–(c) as a set of methodological prescriptions; they do not believe that psychologists should say something about the ontological status of mental states. According to the methodological view, psychologists should aim to scientifically describe and explain behavior without referring to mental states, images, or processes. Radical behaviorists, on the other hand, deny that mental entities exist and argue that private events should be included in the analysis of behavior—that is, that private events should be analyzed in terms of the same principles that have been used to study overt behavior (Day, 1983; Ringen, 1999; Skinner, 1974).

Although behaviorism is generally viewed as a distinctively American psychology, most historians recognize that its roots can be traced back to the work of the Russian objective psychologists or reflexologists (Boakes, 1984; Fuchs & Milar, 2003; Hergenhahn, 2005).³ Ivan Sechenov, often credited as the founder of this school, worked on the (excitatory and inhibitory) mediational role of the cerebral cortex in reflex actions and extrapolated his findings to the concepts of psychology. In *Reflexes of the Brain*, Sechenov (1863/1965) defended the view that every mental process is reducible to a physiological reflex: “only physiology holds the key to the scientific analysis of psychical phenomena” (p. 351). Ivan Pavlov and Vladimir Bekhterev extended Sechenov’s work by independently discovering the principles of classical conditioning. Indeed, Sechenov’s call for an objective psychology seems to have played an important role in Pavlov’s conclusion that

² We include the qualification “in psychology” because the present overview excludes the role behaviorist theories played outside psychology (e.g. philosophy, economics, and sociology). See, for example, Pooley and Solovey (2010) and Hauser (2015).

³ In addition, historians generally recognize that behaviorism also affected the development of psychology outside the United States. See, for example, Ardilla (2009).

his experimental results (dogs salivate in response to food stimuli at a distance) could be explained without positing “psychic secretions” (Pavlov, 1923/1928, p. 39).

Textbooks generally describe the rise of behaviorism in American psychology as a distinctively Watsonian revolution (Todd, 1994). Indeed, Watson (1913) coined the term *behaviorism* and played a substantial role in shaping the debate about objective psychology in his manifesto, “Psychology as the Behaviorist Views It.” Historians, however, offer a more nuanced view, showing that Watson’s manifesto did not receive a lot of attention until the early 1920s (Coleman, 1988; Leahey, 2001; Samelson, 1981), when psychologists like W. S. Hunter, J. R. Kantor, Zing-Yan Kuo, Max Meyer, and A. P. Weiss started to defend distinctively behaviorist conceptions of psychology (Mills, 2012, p. 101). Historians also note that Watson was by no means the first to practice objective psychology in the American psychological community. Psychologists like E. L. Thorndike—who, despite discovering the law of effect, never identified as a behaviorist nor was viewed as such by his peers—William James, Jacques Loeb, and Knight Dunlap also used objective methods prior to 1913 (Burnham, 1968; O’Donnell, 1985; Todd, 1994).

In the late 1920s and early 1930s, the development of behaviorism entered a new phase. Influenced by P. W. Bridgman’s call for operational definitions of theoretical concepts as well as the increasing popularity of logical positivism in the United States (Carroll, 2017; Walter, 1990; Verhaegh, 2019), behaviorists like Hull, Skinner, and Tolman started to develop more rigidly formulated variants of behaviorism; even though their deductive (Hull), inductive (Skinner), and purposive (Tolman) approaches strongly differed from each other.⁴ As a result, most historians distinguish between classical behaviorism and neobehaviorism in their views about the development of mid-20th-century American psychology (DeGrandpré & Buskist, 2000; Parot, 2001; Smith, 1986).

After the Second World War, the popularity of behaviorism gradually declined. The waning influence of logical positivism, the pioneering work of G. A. Miller, Allen Newell, and Herbert Simon, developments in artificial intelligence and linguistics, the invention of the computer (and thereby the growing popularity of a computational perspective on cognition), as well as doubts about the scope of the behaviorist approach (Breland & Breland, 1961; Chomsky, 1959; Lashley, 1951), contributed to what has been called the “cognitive turn” in psychology. Where behaviorists had viewed psychology as a science of behavior, the cognitivists developed new methods, approaches, and frameworks to study “mental” processes in a strictly empirical fashion.

The nature of the cognitive turn has been highly disputed by historians of psychology. Some view the cognitive turn as a paradigmatically Kuhnian revolution (Baars, 1986; Gardner, 1985); others have suggested that there is much continuity between cognitive psychology and methodological behaviorism (Leahey, 2001; Mandler, 2002), or even that the very notion of a “cognitive turn” should be dismissed as an “origin myth”, invented in the early 1970s by cognitive psychologists (Dember, 1974; Joynson, 1970) to foster a shared identity (Hobbs & Chiesa, 2011). Still, historians generally agree that fewer and fewer psychologists identified as behaviorists from the late 1950s onward.

Behaviorism: The Numbers

In this article, we use bibliometric tools to study the development of American psychology between 1920 and 1970. Bibliometrics is a field of study that is characterized by the use of statistics for analyses of written publications such as articles or books. It has proven difficult to pin down exactly when and where the practice of bibliometric

⁴ It should be noted that although Skinner’s early work was heavily influenced by operationism and logical positivism, he started to dismiss these views about the nature of science from the mid-1940s onward. See Skinner (1945a, 1945b), Allen (1980), and Verhaegh (2019).

analysis began (Broadus, 1987; Hood & Wilson, 2001) but historians often cite Eugene Garfield's development of the Science Citation Index in 1955 as the birth of modern bibliometrics (Garfield, 1963). As the available technologies improved during the 1960s and 1970s, bibliometrics developed from a limited field of study to a practice that uses a wide variety of data to accomplish a wide variety of goals (Liao, Zhang, Wang, & Wang, 2016; Martínez-Gómez, 2015). Nowadays, there are bibliometric analyses of disciplines as diverse as history, information science, and pharmacotherapy (Buchanan & Hérubel, 1997; Lee, 2008; Thompson & Walker, 2015).

One of the first bibliometric analyses that aimed to track the development of psychological schools was published in the late 1990s. Robins, Gosling, and Craik (1999) investigated the development of psychoanalysis, behaviorism, cognitive psychology, and neuroscience in the second half of the 20th century to answer the question "[w]hich, if any, of the schools currently competing for intellectual influence and institutional power is most *prominent*" (p. 117, our emphasis). Their assessment of "prominence" consisted of three indexes. First, the authors selected four journals (i.e., *American Psychologist*, *Annual Review of Psychology*, *Psychological Bulletin*, and *Psychological Review*) which they considered to be representative of the entire field of psychology and assessed its contents using the PsycINFO database. Second, they examined the subject matter of dissertations published between 1967 and 1994. Finally, the authors selected four of the top journals of each school (e.g., *Journal of Experimental Analysis of Behavior* for behaviorism) and developed a prominence index by "computing the total number of times per year the flagship publications cited articles published in each subdisciplinary field" (Robins et al., 1999, p. 119). Their results suggested that (a) there is little interaction between psychoanalysis and the rest of psychology, (b) the decreasing popularity of behaviorism gave rise to cognitive psychology, and (c) cognitive psychology was still the most prominent school in the 1990s.

The main methodological challenge in studying the development of a discipline is to develop a reliable way to determine which authors and publications belong to which school. For most schools and approaches, it is extremely difficult to come up with a satisfying definition. Most definitions of behaviorism, for example, will be either (a) so strong that they exclude a whole range of authors that are typically perceived as representatives of the school or (b) so weak that they fail to exclude authors and publications from rival schools. This problem seems to be especially pressing for the classification of behaviorist psychologists and publications because historians have even debated the question to what extent cognitive psychology should be viewed as a variant of methodological behaviorism.

Robins et al. (1999) tried to circumvent such classification problems by using keywords. In determining the number of behaviorist publications, for example, they used the keywords *reinforc#* and *conditioning* to represent behaviorism. They searched for these keywords in the article titles, abstracts, subject indices, and the keyword phrases listed by the authors.⁵ This keyword approach has proven influential. In recent years, many bibliometric studies in the historiography of psychology collected their data using keyword searches and/or similar measurement techniques (e.g., Burman, Green, & Shanker, 2015; Guilera, Barrios, & Gómez-Benito, 2013; Liu & Oakland, 2016; Schui & Krampen, 2010).⁶

In the present study, however, we develop a different approach. Although we will study the occurrence of terms like *reinforc#* and *conditioning* in article titles (more on this below), we

⁵ For psychoanalysis, cognitive psychology and neuroscience, they used the keywords *psychoanal#*, *cognit#*, and *neurosci#* or *neuropsych#*, respectively. They did not use the keywords *behav#* or *behavior#* because they deemed these words too generic.

⁶ There has been a number of articles that looked at specific schools of psychology (Bala & Gupta, 2010; Liu & Oakland, 2016; Schui & Krampen, 2010), but so far there has been no bibliometric study that focuses on the rise and fall of behaviorism.

refrain from using keyword analyses to study the development of American psychology between 1920 and 1970. Our reasons are threefold. First, many articles published between 1920 and 1970 do not have abstracts or author keywords. Our data contain titles and citation data for each of the 119,278 articles published in this period but this is too limited information to solely rely on a keyword analysis. After all, it is implausible that most or even a majority of the behaviorist articles will use terms like *reinforc#* and *conditioning* in their titles.⁷ Second, we believe that keyword analyses should be generally used with caution because it is unclear to what extent they measure what they ought to measure. It is unclear, for example, what proportion of articles containing the term *cognit#* will actually belong to the cognitivist school and, conversely, what proportion of cognitive psychology articles will contain the term *cognit#*. Indeed, Robins et al. (1999) themselves admit that “keyword analyses do not detect all of the articles related to each school” and that they do not expect their “keywords to capture the full range of articles published in the flagship publications” (p. 119n1). Third, we refrain from using keywords to classify authors and publications because most demarcation criteria based on keywords are too rigid. Not only do keyword classifications ignore conceptual changes—for example, the fact that the meaning of concepts like *reinforce#* and *conditioning* changed considerable because of especially Skinner’s theoretical and conceptual innovations from the mid1930s—most keyword analyses also do not allow for the possibility that (a) whether or not an author or a publication should be classified as behavioristic is a matter of degree, and (b) “behaviorism” itself is a tremendously complex, multidimensional concept, such that it is well possible that authors or publications can be behavioristic in one sense but not in another.

Rather than employing unnecessarily inflexible, external standards to determine whether or not a particular author qualifies as a behaviorist, the present study will study the structure and development of American psychology by mapping the *similarities* between authors and concepts. In studying cocitation networks, for example, we do not use a strict demarcation criterion to determine whether or not a psychologist like J. R. Kantor, who developed an “interbehaviorist” approach in the early 1920s, should be classified as a “real” behaviorist. Rather, we use citation data to study the way in which Kantor’s work was perceived to be similar to the work of other psychologists by mapping the cocitations between Kantor and every author cited in American psychology journals between 1920 and 1970.⁸ Moreover, in studying cocitation networks, we also allow for conceptual changes over time by mapping the way in which historical actors perceived authors to be similar. It is possible, for example, that E. G. Boring’s position was perceived to be antithetical to Watson’s approach in the 1910s and 1920s but that the general perception changed when the former started developing operationist approaches to psychology from the mid-1930s onward. If this is the case, it will be reflected by the fact that Boring occupies a different position in a network mapping cocitations from the 1920s than in a network mapping cocitations from the 1940s. Cocitation networks allow us to map these perceived similarities without us having to decide whether or not Boring from the 1930s onward qualified as a (methodological) behaviorist.

In relying on cocitation networks, in sum, we deliberately do not rely on an external, theoretical definition of behaviorism; nor do we presuppose that whether or not a certain author or article qualifies as behavioristic cannot be subject to change. Rather, we rely on

⁷ In addition, the keywords provided by the indexing databases (e.g. PsycINFO; Web of Science) themselves are notoriously unreliable, especially when used to map developments before the 1960s. Burman (2018), for example, shows that throughout the development of the PsycINFO database, the meaning, content, and function of the “metadata” (e.g. keywords) has changed drastically, resulting in different classifications of articles over different time periods. In the 1960s, for example, the American Psychological Association changed its manner of archiving and reporting index terms for both new and old articles to such an extent that all data before the 1960s “simply cannot be used to track trends” (p. 313).

⁸ Co-citation is the frequency with which two documents are cited together. To be strongly co-cited, a large number of scholars must cite the two works together. As such, in measuring co-citation strength, one measures the degree of association as perceived by the population of citing authors (Small, 1976).

a sociological characterization of behaviorism and assess which scholars were considered to be doing similar work by the psychological community at various points in time. The more often two authors are cited together in a particular decade, the closer to each other they will be situated in the cocitation network for that particular decade.

Apart from an analysis on the basis of author cocitation networks, we also employ term co-occurrence networks to describe the structure and development of American psychology. Term co-occurrence networks map the similarities between concepts without presupposing a strict criterion to determine whether or not concepts should be classified as behavioristic. A recent article by Flis and van Eck (2018) describes a similar analysis of 20th-century psychology on the basis of 673,393 articles taken from 1,269 (psychology) journals focusing on the time period 1950 to 1999 to test Cronbach's (1957, 1975) distinction between correlational and experimental psychology and to see whether there were significant changes in both content and amount of output. The authors generated six term maps and identified different clusters of terms representing different fields within psychology. The six term maps showed significant changes in both content and amount of input throughout the time period. In our analysis, we will build upon Flis and Van Eck's work and apply their method to our dataset. This second analysis will provide us with an additional method to assess the structure and development of American psychology. Because our co-occurrence analyses do not depend on citation data, it can be used as an independent check on the validity of our results in the first analysis. Furthermore, we show that some of the differences between the author and the term maps give rise to some interesting hypotheses about the development of 20th-century American psychology.

Method

Data

The publication data were retrieved from Web of Science (WoS). The search was done using the Science Citation Index Expanded (SCIE) and the Social Science Citation Index (SSCI) of WoS, the only citation indices that include publications during the time span of interest. An advanced search query using the term *wc=(psychology) OR su=(psychology)* (where "wc" is the WoS category tag and "su" is the research area tag) was run with the three additional requirements that the results should be (a) articles, (b) published between 1920 and 1970, and (c) written in English. This query resulted in 119,429 articles from 246 different journals. These were downloaded with full record and cited references.

Next, the dataset was cleaned by removing all duplicate articles based on their following attributes: author, title, publication name, cited references, and International Standard Serial Number. This led to a total of 119,278 articles in the dataset. Figure 1 shows the 25 journals with the most articles in the dataset, which together contribute 52% of all data. The titles suggest that the major journals in the dataset were virtually all psychology journals. Because the data were not equally distributed over all the decades, the data were analyzed per decade. This ensures that the results are not skewed to the later decades, for which there are many more publications and citations. The number publications in the dataset per decade is shown in Figure 2. This figure shows that, except for 1941–1950, the number of publications grows exponentially every decade.⁹

Coverage of the dataset. Concerns have been raised about the suitability of WoS for historical research. Its coverage of some areas of psychology has been characterized as uneven, containing fewer journals in the areas of organizational and clinical psychology

⁹ Note that the period 1920–1930 consists of 11 years. This is because the overall dataset actually spans 71 years, not 70. We chose to append this extra year to the earliest period to somewhat offset the sparsity of publications in that decade.

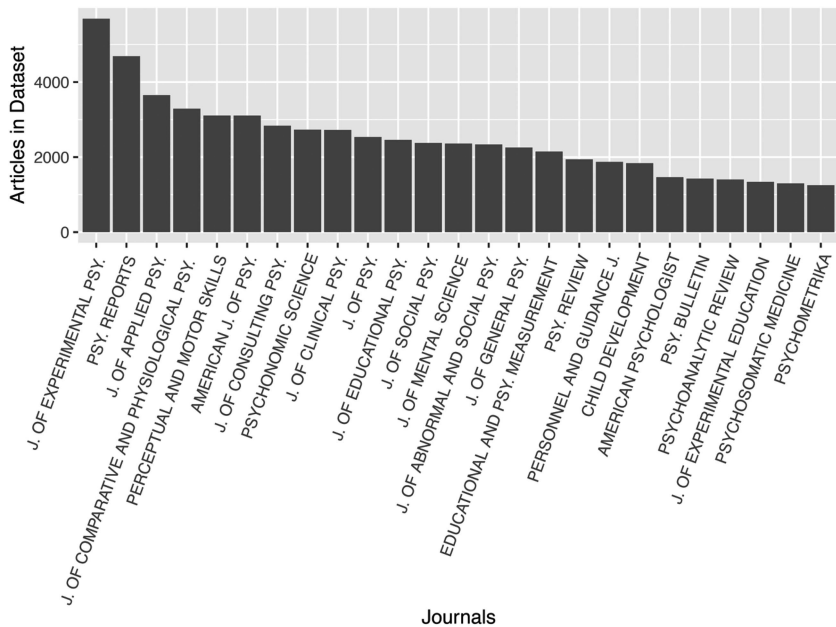


Figure 1. Number of articles per journal for the 25 journals with the most publications in the dataset.

than PsycINFO, while having more extensive coverage of neuroscience, methodology, and statistics journals (García-Pérez, 2010). Approximately 90% of all documents in WoS originates from English-speaking countries (Orduña-Malea & López-Cózar, 2014). While this may reflect a geographic or linguistic bias, this is not particularly troubling for our purposes, as we focus on developments within American psychology. Furthermore, 90% of all documents in WoS are journal articles, which means that coverage of research monographs and conference communications is probably limited (López-Cózar, Orduna-



Figure 2. Number of publications in the dataset for each decade during 1920–1970.

Malea, & Martín-Martín, 2019). This does not preclude citations of research monographs in journal articles from the database, but it might result in underrepresentation of fields where books are a vital aspect of the publication culture (García-Pérez, 2010).

To assess the suitability of the SCIE and SSCI indexes in WoS for our research, we used a list of 15 journals that were regularly used by psychologists in the 1920s and 1930s.¹⁰ WoS nominally contained full records for 13 of these journals for the entire period of 1920–1970 (or from a later starting date if the journal was founded after 1920); two journals were included for a subset of that period.¹¹ By comparison, PsycINFO contained records for nine of the 15 journals for the entire period of 1920–1970; two journals were not included and four journals were included for a subset of that period.¹²

Our evaluation of SCIE and SSCI against our own list of sources suggests that WoS's coverage of major psychology journals is strong. Still, it is possible that coverage for other journals is less comprehensive, and that coverage at lower levels is spotty (e.g., missing issues for included journals, missing articles for included issues, and missing cited references for included articles). To assess in more detail whether the dataset provides full and accurate coverage of the literature from 1920 to 1970, we selected two journals for each decade. The sampling procedure for the journals was as follows. First, for each decade, we ranked the journals in our dataset according to the ratio of cited references to published articles. The first journal was randomly sampled from the bottom 5% of this ranking—thus, the set of journals containing fewer cited references than would be expected based on the average citation practice for that decade. The rationale was that WoS would be more likely to contain incomplete records (e.g., missing or incorrect entries in the cited references column) for these journals. The second journal was randomly sampled from the remaining 95% of journals for that decade.

For each of the 10 selected journals, we checked the issues included in WoS for that decade against the issues as listed in the printed journal itself, or, if that was not available, against the table of contents on the publisher's website. Next, we randomly sampled one issue from each of these journals and checked the articles included in WoS against the articles as listed on the publisher's website. Finally, for five randomly selected articles from the issue, we checked the cited references in WoS against the cited references in the full text. The results of this procedure are displayed in Table 1.

For the journals with a low cited references-to-articles ratio, coverage in WoS varied somewhat between decades. For the journal selected for 1920–1930, one issue was not included, and two out of 20 articles were missing for the selected issue, without any apparent reason (i.e., the missing issue and articles appeared to be regular research

¹⁰ These journals were: *Acta Psychologica*, *American Journal of Psychology*, *American Psychologist*, *Contemporary Psychology*, *Journal of Comparative Psychology* (and its successor *Journal of Comparative and Physiological Psychology*), *Journal of Experimental Psychology* (and its successors *JEP: General*, *JEP: Human Perception and Performance*, and *JEP: Learning, Memory, and Cognition*), *Journal of General Psychology*, *Journal of the Experimental Analysis of Behavior*, *Proceedings of the American Academy of Arts and Sciences*, *Proceedings of the National Academy of Sciences*, *Psychological Bulletin*, *Psychological Review*, *Quarterly Journal of Experimental Psychology*, *Science*, and *The Psychological Record*.

¹¹ The two journals that were not completely included are *Proceedings of the American Academy of Arts and Sciences* and *The Psychological Record*.

¹² The two journals that were not included are *Contemporary Psychology* and *Proceedings of the American Academy of Arts and Sciences*; the four journals that were not completely included are *Acta Psychologica*, *Proceedings of the National Academy of Sciences*, *Quarterly Journal of Experimental Psychology*, and *Science*.

Table 1
Coverage of Web of Science for Randomly Selected Journals in Our Dataset

Decade	Journal	Issue coverage for decade	Article coverage for issue	Cited reference coverage
Bottom 5%				
1920–1930	<i>National Vocational Guidance Association Bulletin/Vocational Guidance Magazine</i>	1 Missing (66/67)	1924–5, 2 Missing (18/20)	Complete
1931–1940	<i>The Human Factor</i>	Complete (55/55) ^a	1932–8, Complete (5/5)	Complete
1941–1950	<i>Occupations: The Vocational Guidance Magazine/Journal</i>	Complete (81/80) ^b	1943–6, Complete (10/10)	Complete
1951–1960	<i>Canadian Psychologist</i>	Only 1960 (4/40)	1960–4, Complete (9/9)	Complete
1961–1970	<i>Personnel Journal</i>	Complete (113/113)	1963–4, 13 Missing (5/18) ^c	Complete
Other 95%				
1920–1930	<i>Journal of Educational Psychology</i>	Complete (11/11)	1921–0, Complete (35/35)	Complete
1931–1940	<i>Journal of Applied Psychology</i>	Complete (60/60)	1933–4, Complete (12/12)	Complete
1941–1950	<i>Journal of Applied Psychology</i>	Complete (60/60)	1946–4, Complete (12/12)	Complete
1951–1960	<i>Educational and Psychological Measurement</i>	Complete (40/40)	1956–4, Complete (14/14)	Complete
1961–1970	<i>Psychological Review</i>	Complete (58/58)	1966–2, Complete (5/5)	Complete

^a This journal was published from 1932 to 1937. It had 12 issues per year. A recurring issue is the annual report and statement of accounts (Issue 3 in 1932 and 1933, Issue 12 in 1934–1936). These annual reports are not included in Web of Science. ^b Web of Science has records for Issue 21(8), which was absent from the publisher's website. ^c The contributions not present in Web of Science all seem to be from a category other than "Original Article," for example, commentaries (sometimes less than a column), book reviews, and so on.

publications). For the journal selected for 1951–1960, only four issues out of many were available, apparently because the journal is only indexed from 1960 onward. For all other decades, WoS contained all issues. A number of articles for the selected issue were missing for 1961–1970. It should be noted, however, that these missing articles were not standard research reports, but rather a variety of very short (sometimes less than half a page) contributions. Finally, the cited references in WoS were complete for each article for each journal. This means that the low ratio of cited references to articles for these journals was not an artifact, but that these articles actually contained very few cited references. For the journals sampled from the remaining 95% WoS included all issues, all articles, and all cited references. Thus, although WoS does not have perfect coverage for some journals, its overall coverage, especially for major psychology journals, was rather impressive and at least satisfactory for our purposes.

Citation data. From our data sets, the 200 most-cited authors per decade were extracted. These were needed for the generation of cocitation networks. The extraction of the top 200 most-cited authors per decade was done using VOSviewer (Version 1.6.7, [van Eck & Waltman, 2010](#)), which extracts and counts the citations for each author.¹³ It is important to note that the cited articles do not have to be from the same decade as the publication data. Although William James died in 1910, for example, he is still one of the top 200 most-cited authors in the 1940s. We extracted a number of authors as close to 200 as possible (extracting more than 200 in case authors were tied for the rank of 200). [Table 2](#) shows the number of authors per decade and the minimum required number of citations. This table shows that the minimum number of citations for an author to belong to the top 200 increases exponentially over the decades.

Author Cocitation Networks

For each decade we created an author cocitation network using the program VOSviewer. The association strength ([Van Eck & Waltman, 2009](#)) between authors was used as the measure of similarity. For a given pair of authors, the association strength is proportional to the ratio between (a) the observed number of cocitations and (b) the expected number of cocitations if they would be statistically independent. Thus, authors with a large number of citations require a large number of cocitations to have a high association strength.

To map the association strengths for all pairs of authors onto a two-dimensional space, the VOSviewer software uses the VOS (visualization of similarities) method, which is similar to the more widely known technique of multidimensional scaling. In addition, VOSviewer uses this method to cluster the authors. In this case, a cluster is a group of authors with high mutual association strengths. The default clustering resolution of 1.0 was used in every network.¹⁴ For a more technical explanation of the clustering technique, see [Waltman, Van Eck, and Noyons \(2010\)](#).

¹³ Some authors appear in the dataset under different names. These different names were normalized as much as possible, but in some cases it was not possible to determine which author a name refers to. For example, the dataset includes the authors “D. G. Brown,” “J. S. Brown” and “Brown.” It is then not possible to assign the citations of “Brown” to either D. G. or J. S. Brown. In this case, “Brown” was retained as a separate author.

¹⁴ While 1.0 is the default value, this does not in any way represent a “neutral” value. The optimal clustering resolution may be different for each dataset. A common recommendation is to explore different values for each map and choose the one that yields the most satisfactory result. We believe this approach entails a risk of confirmation bias, however, because the result that is deemed most satisfactory is likely to be the result that matches the researcher’s preconceptions. Still, we did explore different resolution values, but only to see whether the value of 1.0 represents a stable solution. We did not see appreciable differences for different parameter values (ranging from 0.8 to 1.2), unless reported otherwise.

Table 2
Number of Authors Included in the Dataset and Their Publication Threshold per Decade

Decade	Number of authors	Minimum number of citations
1920–1930	201	24
1931–1940	200	63
1941–1950	202	74
1951–1960	202	150
1961–1970	200	342

Term Co-Occurrence Networks

To create the term co-occurrence networks, we used the data from all 119,278 publications. Using the VOSviewer software, the titles¹⁵ were extracted from the publication data. The software then identified and extracted the longest possible noun phrases (or terms) from the titles. Occurrences and co-occurrences of the terms were counted using a binary counting method, such that multiple occurrences of a term in a given title were counted as one occurrence. From the list of terms for each decade, the 100 most frequent terms were selected.¹⁶ We used a fixed number of terms to keep the networks comparable across decades. For the 100 most frequent terms per decade, association strengths were computed in the same way as for the authors. To create the term networks, VOSviewer uses the same mapping and clustering algorithm as for the author cocitation networks.

Results

Cocitation Networks

Cocitation networks per decade. Table 2 provides an overview of the number of clusters generated by VOSviewer for each decade and the number of authors that are placed within these clusters. The clusters displayed in boldface are the clusters in which authors who are typically classified as behaviorists were most strongly represented. For each decade, we will describe the cocitation network in relation to the information presented in Table 3.

1920–1930. For the decade 1920–1930, VOSviewer generated five clusters, shown in Figure 3. Cluster 3 (blue) contains most authors who are typically viewed as behaviorists by historians (e.g., the above-mentioned Watson, Tolman, Hunter, Kuo, and Weiss;

¹⁵ We relied on titles because full texts or even abstracts were available only for a subset of the articles. This requires that titles be representative of the contents of the articles for our conclusions to be valid. Given the strongly descriptive and matter-of-fact nature of many of the titles in the dataset, we believe the data go a long way toward meeting this assumption. Moreover, sporadic instances of titles containing terms that have little bearing on the contents of the article, or of titles lacking essential information regarding the object or method of study, only have a minor influence on the overall picture, because each network is based on thousands of titles. That said, an analysis based on full texts would most likely yield even richer, more nuanced conclusions.

¹⁶ VOSviewer also computes relevance scores for terms, such that frequent but uninteresting terms (e.g. *study*) may be filtered out. We refrained from using relevance scores as this would intermittently leave out terms like *behavior*, thereby possibly distorting the results by removing terms associated with behaviorism. The only term we excluded from the networks (1931–1940; 1941–1950) was *iii*, which was erroneously detected as a noun phrase.

Table 3
The Number of Citations and Authors for Each Cluster per Decade

Decade	Cluster					
	1 (Red)	2 (Green)	3 (Blue)	4 (Yellow)	5 (Purple)	6 (Aqua)
1920–1930						
Citations	3,994	3,149	2,280	2,156	640	
Authors	64	51	38	36	12	
1931–1940						
Citations	10,373	7,562	5,363	3,000	1,931	
Authors	75	63	34	16	12	
1941–1950						
Citations	10,073	9,157	5,892	4,109	2,385	835
Authors	67	56	33	24	15	7
1951–1960						
Citations	28,915	18,696	14,807	7,848		
Authors	86	55	35	26		
1961–1970						
Citations	44,779	35,549	28,388	15,656	6,905	
Authors	69	55	46	17	13	

Note. Boldface indicates the clusters in which authors who are typically classified as behaviorists were most strongly represented.

but also H. C Warren, W. T. Heron, and H. Cason).¹⁷ The cluster also contains quite a few biologists and physiologists whose work was often appealed to by behaviorists (e.g., C. P. Stone, G. E. Coghill, W. T. Preyer, and W. J. Crozier). The most-cited authors in Cluster 3 are Watson (234 citations) and Hunter (178 citations).

If we interpret Cluster 3 as the “behaviorist cluster,” then behaviorist authors make up 19% of the top 201 most-cited authors of 1920–1930, and also account for 19% of their citations. As such it is the third largest cluster. Cluster 1 is substantially larger, accounting for 32% of the most-cited authors and 33% of the citations between 1920 and 1930. Cluster 1 does not contain any author who is typically classified as a behaviorist¹⁸ and contains mostly psychologists who developed the field of mental testing and the closely related field of educational psychology (e.g., L. L. Thurstone, L. M. Terman, A. Binet, and C. E. Spearman, and G. M. Whipple). The most-cited authors of Cluster 1 are Thorndike (331 citations) and Terman (186 citations).¹⁹

Cluster 2 is also substantially larger than the “behaviorist cluster” (25% of the most-cited authors and 26% of the citations) and contains mostly psychologists connected to Gestalt psychology (e.g., K. Koffka, M. Wertheimer, and W. Köhler) and the structuralist school (e.g., E. B. Titchener and E. G. Boring). The most-cited authors of Cluster 2 are Titchener (372 citations), C. E. Seashore (141 citations), and Boring (131 citations). The most-cited authors overall are Titchener and Freud (359 citations). Freud’s position

¹⁷ Not all names mentioned are visible in Figure 3, which only displays the most-cited names.
¹⁸ The only exception is Clark Hull, who was mostly working on aptitude testing in the 1920s. It was only in the late 1920s that Hull started to develop a neobehavioristic explanation of higher mental processes.
¹⁹ Despite the fact that he discovered the law of effect, Thorndike, as mentioned above, never identified as a behaviorist nor was viewed as such by his peers. Our analysis of the citation data underlying the co-citation network in Figure 3 confirms this view as it shows that especially Thorndike’s (1921) *Teacher’s Word Book* was frequently cited.

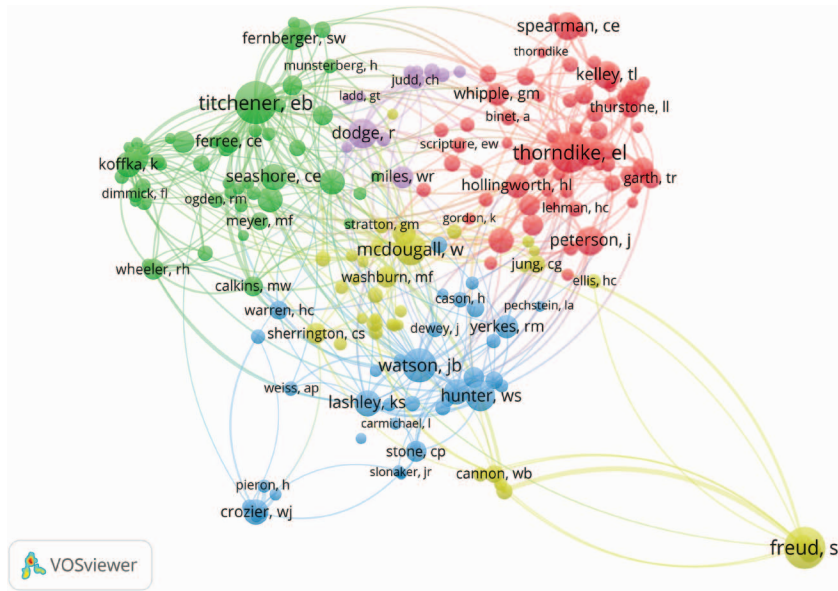


Figure 3. Cocitation network of the 201 most cited authors from the period 1920–1930. See the online article for the color version of this figure.

in the cocitation network shows that there was little interaction between psychoanalysis and the rest of psychology (thereby contextualizing Robins et al.'s (1999) conclusion about the development of psychoanalysis after the Second World War).

1931–1940. For the decade 1931–1940, VOSviewer generated five clusters, shown in Figure 4. Cluster 3 (blue) contained the largest number of authors who are typically classified as behaviorists (e.g., Tolman, Hunter, Watson, Kuo, Spence, and Weiss), yet Cluster 4 (yellow) also contains a substantial share of authors who are typically viewed as behaviorists (e.g., Hull, Pavlov, Skinner, Razran, and Guthrie).²⁰ If we view these clusters together as a “behaviorist supercluster,” then behaviorist authors make up 25% of the top 200 cited authors of 1931–1940, and account for 30% of the citations. The most-cited authors included in Clusters 3 and 4 are Hull (593 citations), K. S. Lashley (583 citations), and Tolman (419 citations).

It is tempting to attribute the appearance of two “behaviorist clusters”²¹ to the contemporaneous emergence of “neobehaviorism,” a movement generally associated with Skinner, Hull, and Tolman, and characterized by a stronger focus on formalizing the laws of behavior. Such an interpretation is not without problems, however; whereas Hull and Skinner are in the same cluster, Tolman is not. Still, it might be possible that the two clusters track a distinction between older and younger writers. Tolman (despite the fact that he would later be classified as a neobehaviorist), Watson, Spence, Kuo, and Muen-

²⁰ Pavlov's more prominent presence in this network compared to that of 1920–1930, is because of the publication, in 1927, of the English translation of *Conditioned Reflexes: An Investigation of the Physiological Activity of the Cerebral Cortex* (Pavlov, 1927).

²¹ To ensure this is not an artefact of the clustering resolution parameter value of 1.0, we tried coarser parameter values, until the number of clusters was reduced to four. Even in this map, the two “behaviorist clusters” remained separate (while the green and yellow clusters from Figure 4 had merged). Thus, these two subclusters most likely capture a real structure in the underlying data.

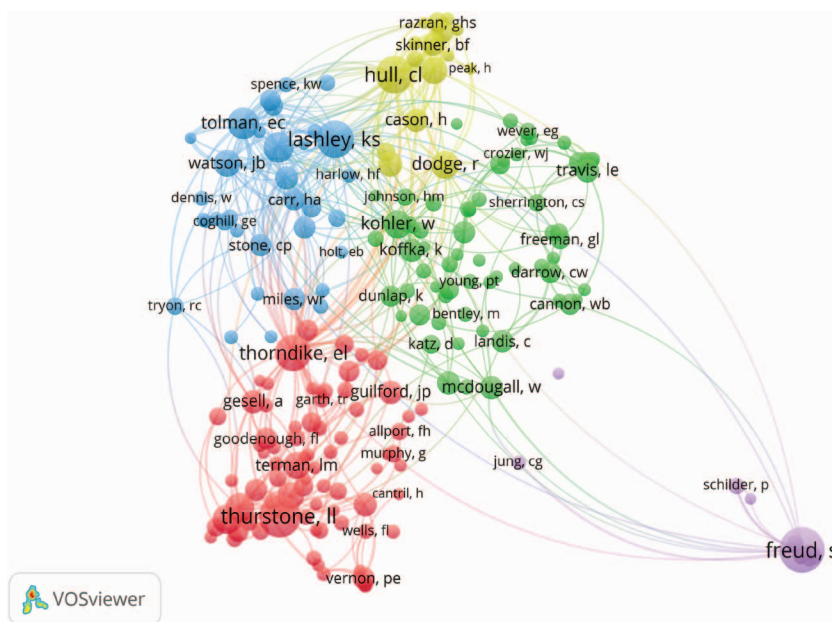


Figure 4. Cocitation network of the 200 most cited authors from the period 1931–1940. See the online article for the color version of this figure.

zinger all started publishing well before 1930 and/or had a background in comparative psychology, whereas the behaviorists in the yellow cluster started publishing later (Skinner, Razran), were translated into English later (Pavlov), or were doing little experimental work with animals in the 1920s (Hull, Guthrie).²²

All the same, even combined, this “behaviorist supercluster” does not constitute the largest cluster within psychology. Both Cluster 1 (red) and Cluster 2 (green) individually are larger than the two “behaviorist clusters” combined. The authors in Cluster 1 account for 37% of the citations and the cluster contains mostly psychologists connected to the mental testing literature (Thurstone, Thorndike, and Spearman are the most-cited authors). Cluster 2 contains 32% of the most-cited authors and consists mostly of authors connected to Gestalt psychology and the structuralist school (the most-cited authors of the cluster are Köhler, L. E. Travis, and Koffka). Overall, Freud (868 citations) is the most-cited author in the 1930s.

1941–1950. For the decade 1941–1950, VOSviewer generated six clusters, shown in Figure 5. Virtually all authors who are typically classified as behaviorists (e.g., Hull, Tolman, Spence, Pavlov, Skinner, Hunter, Guthrie, Razran, Watson) are located in Cluster 2 (green). The most-cited members of this cluster are Hull (829 citations), Thorndike (400 citations), and Tolman (368 citations). If we interpret this as the “behaviorist” cluster, then behaviorist authors make up 28% of the top 202 most-cited authors of 1931–1940 and account for 28% of their citations. As such, this cluster is only slightly smaller than Cluster 1, which contains 33% of the most-cited authors and 31% of the citations. The

²² We thank François Tonneau (personal correspondence) for this suggestion.

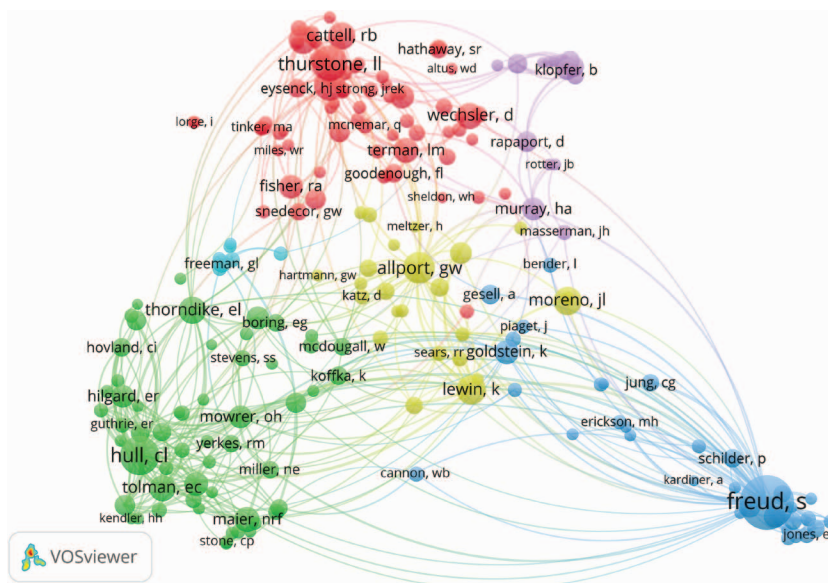


Figure 5. Cocitation network of the 202 most cited authors from the period 1941–1950. See the online article for the color version of this figure.

most-cited authors in this cluster are Thurstone (696 citations), J. P. Guilford (546 citations), and C. Burt (425 citations), all connected to the study and testing of human intelligence. Overall, Freud (1,612 citations) is, again, the most-cited author.

To nuance the suggestion that Cluster 2 can be identified as a “behaviorist cluster,” we should mention that the cluster also contains a large share of authors who are not typically classified as behaviorists. The cluster contains several Gestalt psychologists (e.g., Köhler and Koffka) and functionalists (e.g., R. S. Woodworth and W. James).²³ A possible explanation for this composition is that the cluster does not so much represent a coherent behaviorist movement as a collection of authors who published on a common topic. This means that the cluster may not only contain proponents of behaviorism, but opponents too. Publications that are critical about behavioristic assumptions or findings by behaviorist authors may well end up being cocited with the publications they discuss. Conversely, some authors may be included in this cluster because they were critically cited in behavioristic publications.

1951–1960. For the decade 1951–1960, VOSviewer generated four clusters, shown in Figure 6. Again, virtually all authors who are typically classified as behaviorists (e.g., Hull, Spence, Skinner, Tolman, Estes, Pavlov, and Guthrie) are located in Cluster 2 (green). The most-cited authors of Cluster 2 are Hull (1,538 citations) and Spence (1,077 citations). Somewhat surprisingly, Skinner (600 citations) is not even represented in the top-5 most-cited authors of Cluster 2, despite the received view that Skinner came to dominate behaviorist research after the Second World War (see footnote 2). If we interpret Cluster 2 as the “behaviorist” cluster, then behaviorist authors make up 27% of the top 202

²³ The cluster also contains operationist psychologists like E. G. Boring and S. S. Stevens, but this only confirms the above-mentioned hypothesis that operationism was generally viewed as a variant of (methodological) behaviorism.

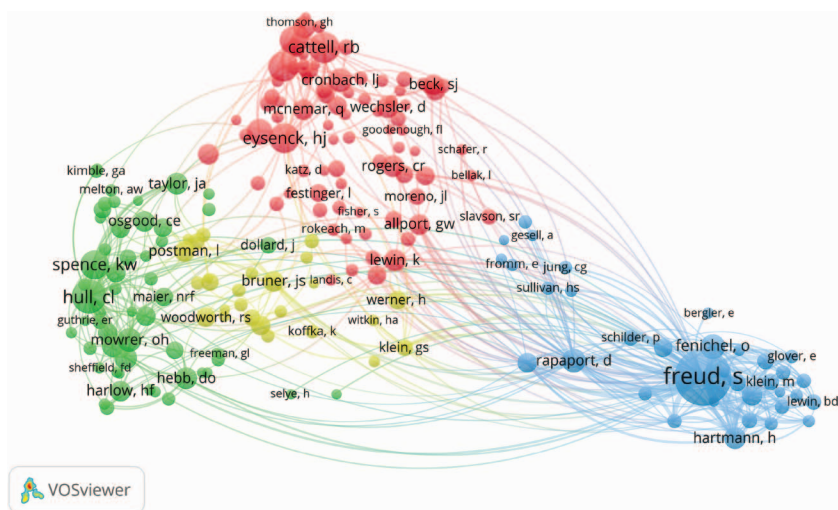


Figure 6. Cocitation network of the 202 most cited authors from the period 1951–1960. See the online article for the color version of this figure.

most-cited authors and account for 27% of their citations. Cluster 1 is substantially larger, accounting for 43% of the most-cited authors and 41% of the citations between 1951 and 1960. Cluster 1 does not contain any author who is typically classified as a behaviorist. The most-cited authors of Cluster 1 are R. B. Cattell (1,264 citations), Guilford (1,191 citations), and H. J. Eysenck (1,167 citations).

The increasing gap between Clusters 1 and 2, compared to the cocitation network for the period 1941–1950, is explained by the fact that the Gestalt psychologists, the functionalists, and the operationist psychologists (Koffka, Köhler, Wertheimer, James, Woodworth, Boring, and Stevens) are now located in a separate cluster (Cluster 4, yellow). In general, the structure of the map shows a greater dispersion between clusters in comparison to the first two decades. Most of the “behaviorist” cluster is located at the bottom left corner of the map, more remote from other clusters than in 1941–1950. The density of connections within this cluster and the relative sparsity of connections to other clusters suggest that behaviorism, although sustaining a substantial number of citations and authors, started occupying a slightly more peripheral position within American psychology.

1961–1970. For the decade 1961–1970, VOSviewer generated five clusters, shown in Figure 7. Again, virtually all authors typically classified as behaviorists are located in Cluster 2 (green). The most-cited author is Spence (1,603 citations); Skinner is still not included in the top five of most-cited authors in his cluster. If we interpret Cluster 2 as the behaviorist cluster, it remains the second-largest cluster with 28% of the most-cited authors and 27% of the citations. Cluster 1 is substantially larger, with 35% of the most-cited authors. Its most-cited authors are Eysenck (2,348 citations), Cattell (2,125 citations), and A. L. Edwards (2,048 citations). Freud (6,290 citations) remains the most-cited author overall.

Notably, Cluster 3 (blue) represents the emerging field of cognitive psychology, including authors such as G. A. Miller, N. Chomsky, G. H. Bower, and B. J. Underwood. Although this cluster still appears to be slightly smaller than the “behaviorist” cluster

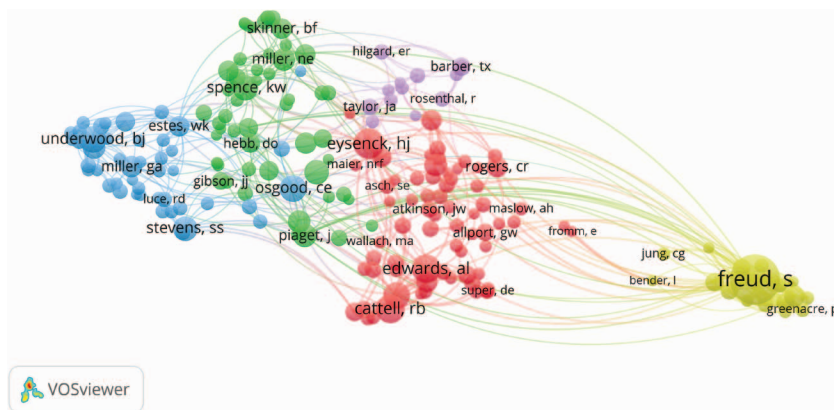


Figure 7. Cocitation network of the 200 most cited authors from the period 1961–1970. See the online article for the color version of this figure.

(23% vs. 28% of the most-cited authors; 22% vs. 27% of the total citations), it should be noted that Cluster 2 also contains a few cognitive psychologists and developmental psychologists (e.g., J. S. Bruner and J. Piaget) who are not typically classified as behaviorists.

Compared to the previous two decades, the “behaviorist cluster” occupies a more central position on the map. This may partly be explained by the relation between behaviorism and cognitive psychology: in both subfields, the same phenomena (e.g., memory, learning) are studied, albeit from a different set of epistemological and methodological assumptions. One of the catalysts of the “cognitive turn” was a perceived inadequacy of behaviorism to satisfactorily explain these phenomena. It is therefore not surprising to find many cocitations between these clusters. The publications concern similar subject matter and authors affiliated with the cognitive revolution often defined their positions relative to behaviorism.

Summary of the cocitation networks per decade. In none of the decades that we analyzed did authors who are typically classified as behaviorists belong to the most prominent (combination of) clusters. This runs counter to the view that psychology was dominated by behaviorism in this part of the 20th century. Even on a very catholic definition of behaviorism—combining the citations of all the clusters that contain authors who are typically classified as behaviorists—the proportion of behaviorists was never larger than 28% of the most-cited authors in a particular decade.²⁴ Importantly, in none of the decades that we studied did the largest cluster contain authors who are typically classified as behaviorists. Moreover, our analysis also rules out an interpretation in which behaviorism may have been a small movement in an absolute sense, but larger than any other cluster in an otherwise disorganized field. In all decades, measurement of personality and mental abilities (championed by authors such as Cattell, Eysenck, Thurstone, Guil-

²⁴ We call this a very catholic definition of behaviorism because we have shown that these “behaviorist clusters” regularly also contained authors who are typically classified as Gestalt psychologists, operationists, functional psychologists, developmental psychologists, and even cognitive psychologists.

ford, Burt, and Edwards) seems to be the main preoccupation of the field. Moreover, in the 1920s and 1930s, the clusters that appear to consist predominantly of Gestalt psychologists (Köhler, Koffka, Lewin) generated more citations than behaviorism.

On the other hand, our cocitation analyses also show that the influence of behaviorism did not dramatically decline after the Second World War either (at least up until 1970). In the 1960s, the behaviorist cluster still made up 28% of the field, more or less the same percentage as in the 1940s and 1950s. Absolutely speaking, the behaviorist cluster(s) even doubled in size almost every decade (except for the 1940s), growing from 8,363 citations in the 1930s, to 18,696 citations in the 1950s, to 35,539 citations in the 1960s. These results suggest not only that the share of citations to behaviorist articles remained stable between 1940 and 1970, but also that the absolute number of citations to the most important behaviorists quadrupled in the same period.

Term Co-Occurrence Networks

1920–1930. For the decade 1920–1930, VOSviewer generated seven clusters, which are shown in Figure 8. The aqua cluster contains terms ($n = 9$) that are typically associated with behaviorist research, such as *behavior*, *white rat*, and *maze*, whereas no such words occur in other clusters. The other clusters roughly correspond to areas such as clinical (yellow), educational (orange), and perceptual psychology (blue). The largest cluster (red) seems to contain mostly general terms, such as *psychology*, *theory*, and *study*.

There is no clear spatial separation between the “behaviorist” cluster and some of the other clusters; in fact, it falls almost completely within the boundaries of the larger green and purple clusters. Thus, it seems that this clustering is somewhat forced onto the data and that a given term’s cluster membership does not imply it does not co-occur with terms from other clusters. Indeed, if we decrease the clustering resolution from 1.0 to 0.8, then

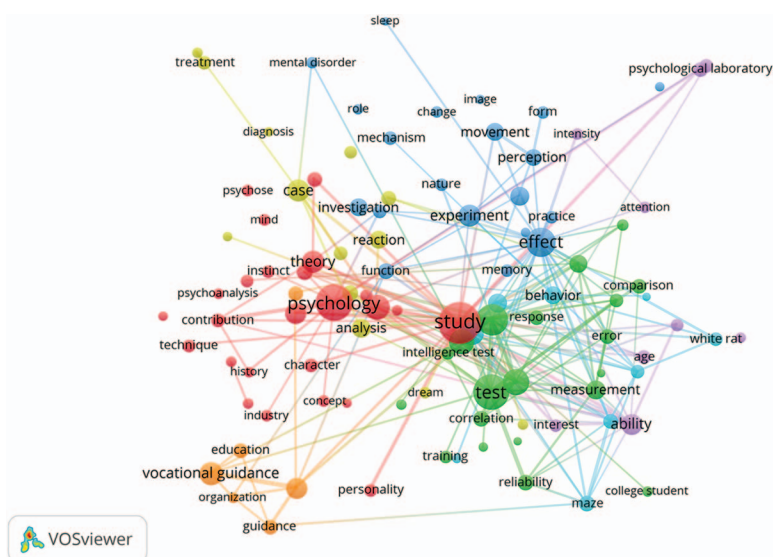


Figure 8. Term co-occurrence network for 1920–1930. See the online article for the color version of this figure.

the “behaviorist” cluster merges with the blue cluster (which still intersects the green cluster). This new cluster ($n = 24$) may best be described as “experimental psychology” (containing terms such as *experiment*, *perception*, and *memory*).

All in all, the term co-occurrence map for this decade shows that a behaviorist vocabulary can already be distinguished in the publications, but also that the different clusters tend to be quite similar to one another, at least when compared to later decades.

1931–1940. For the decade 1931–1940, VOSviewer generated six clusters, which are shown in Figure 9. Terms typically associated with behaviorist research (*behavior*, *learning*, *discrimination*) are included in the red cluster ($n = 27$), which may best be described as representing experimental psychology. If we set the clustering parameter to a minimally finer value (1.0 to 1.01), this cluster breaks down into several subclusters, of which one remaining subcluster ($n = 12$) consists exclusively of terms one would expect in behaviorist research, and a few typical behaviorist terms (*conditioning*, *reflex*) located in a larger adjacent cluster.

Compared to the network for 1920–1930, the cluster with behaviorist terms is larger and more clearly separated from the other clusters. Looking ahead at the maps that will follow, we might see here the start of a bifurcation that will be most apparent from the 1950s onward: correlational psychology on the left and experimental psychology on the right (for a discussion on how to characterize these two superclusters, see Flis & van Eck, 2018).

1941–1950. For the decade 1941–1950, VOSviewer generated six clusters, which are shown in Figure 10. The yellow cluster ($n = 15$) contains terms typically associated with behaviorist research (*behavior*, *discrimination*, *rat*), but also terms that have a wider usage in experimental psychology (*effect*, *influence*, *degree*). Most of this cluster is clearly separated from the other clusters in the map, although one term (*study*) intersects with

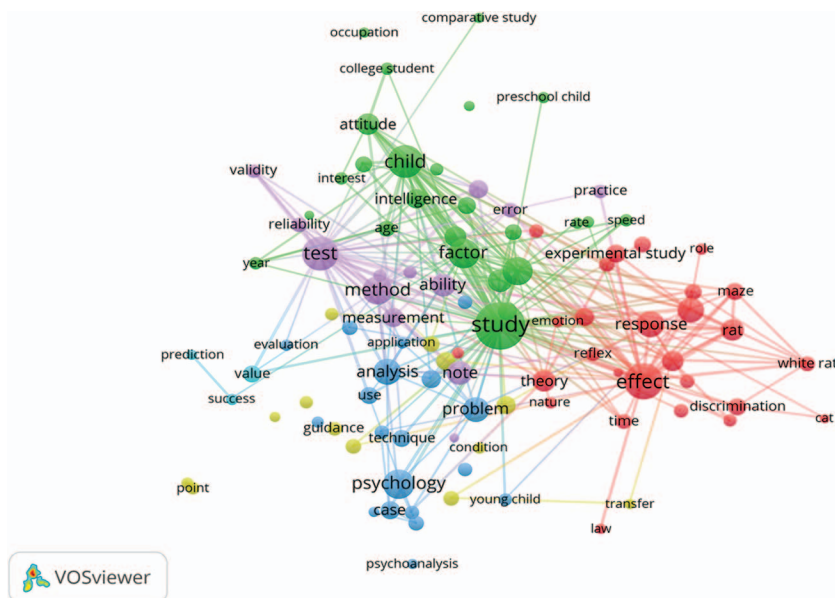


Figure 9. Term co-occurrence network for 1931–1940. See the online article for the color version of this figure.

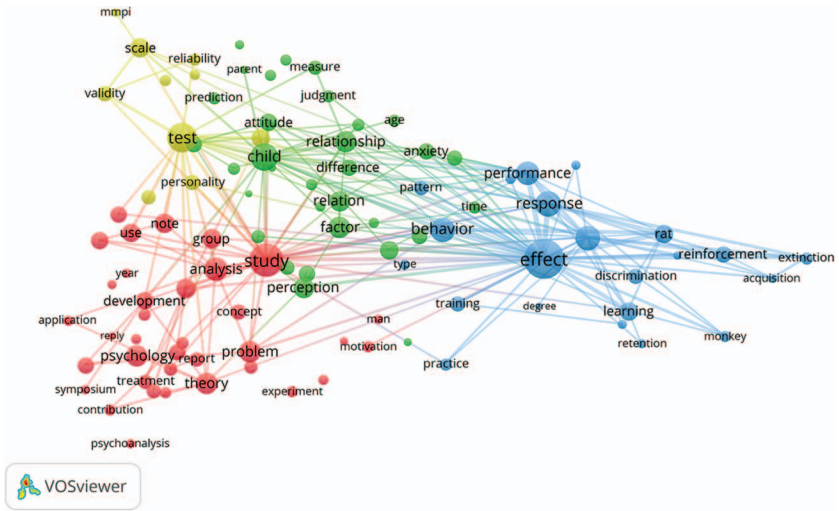


Figure 11. Term co-occurrence network for 1951–1960. See the online article for the color version of this figure.

1961–1970. For the decade 1961–1970, VOSviewer generated four clusters, which are shown in Figure 12. Typical behaviorist terms ($n = 21$) appear in the blue cluster. As in the previous decade, this cluster contains both terms that are highly specific to behaviorism (*reinforcement, extinction, reward, resistance*) and terms that are also used outside of behaviorist contexts (*effect, function, response*). The latter set of terms is again located near the center of the map. Compared to 1951–1960, the number of behaviorist terms remains more or less constant. The yellow cluster contains several novel terms that reflect the emergence of cognitive psychology (*short-term memory, recall, information*).

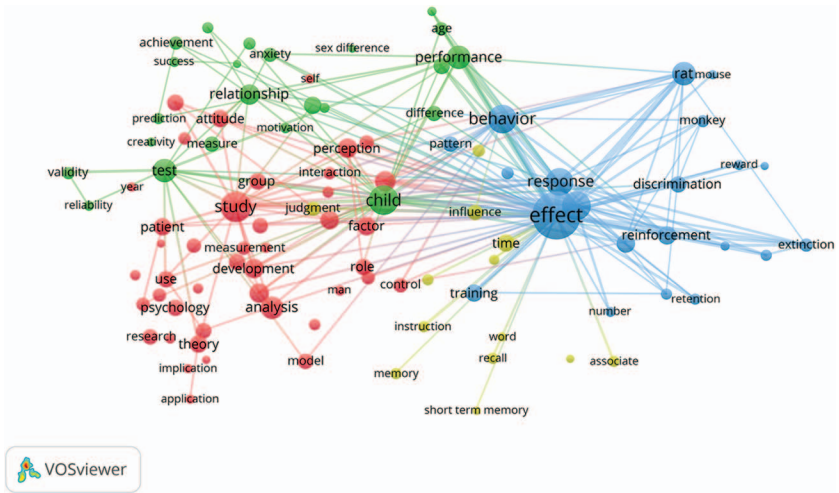


Figure 12. Term co-occurrence network for 1961–1970. See the online article for the color version of this figure.

Summary of the term co-occurrence networks per decade. When we track the development of the term co-occurrence networks across the time period of 1920–1970, we can draw a number of conclusions. First, the networks appear to develop a more orderly structure over time, with smaller numbers of clusters (from seven to four) and less spatial overlap between clusters. This might well reflect the development of the overall field of psychology, in which several subdisciplines became integrated throughout the 20th century. Second, the share of terms typically associated with behaviorist research in the top 100 title words for each decade increases until at least the 1950s. This might be because the behaviorist vocabulary grew over these decades, such that new terms were coined, or rather, that existing terms were increasingly used in a specific behaviorist sense. Alternatively, certain terms, which were already in use earlier, could have become more frequent in titles and entered the top 100 at the cost of other terms. In both scenarios, a plausible explanation is that more publications appeared that can be characterized as behavioristic.

There are some interesting commonalities and differences between the author and term networks. Both series of maps are similar in that they display growing popularity of authors and concepts typically associated with behaviorism after the 1920s, but whereas the size of the behaviorist clusters remained stable in the author networks, the popularity of the behaviorist vocabulary continues to grow (and plateaus in the 1960s). We see two potential explanations. First, it is possible that a lot of behaviorist work was published after the 1940s, but that this was work generated relatively few citations in the field of psychology as a whole. Second, some terms in the “behaviorist clusters” might have been used in work that scholars would not commonly classify as behaviorist, but that did build on methods developed by paradigmatic behaviorists, or was concerned with phenomena (e.g., behavior, learning) that were originally put on the agenda by paradigmatic behaviorists. In that scenario, what these networks might reflect is the success of behaviorists in setting the direction of at least a part of psychological inquiry (cf. Roediger, 2004).

The combination of the term co-occurrence networks and author cocitation networks could also shed light on another discussion. From the term co-occurrence networks alone, one might get the impression that psychology, at least from the 1950s onward, consists of two subdisciplines, characterized as experimental psychology and correlational psychology (Flis & van Eck, 2018), that coexist with very little interaction. The author cocitation analyses, however, suggest a different interpretation. If both subdisciplines were truly separated, one would expect to see this reflected in the author cocitation networks, such that authors should mostly cite work from other authors in their own area. This was not the case. The different overall structure of the author cocitation networks, then, might reflect the simple fact that authors do regularly cite work from those working in different disciplines. For example, it is common for experimental studies on the influence of one variable on another to refer to studies showing correlational evidence for a link between these variables. The cocitation map is also consistent with a more radical reinterpretation, however; the term co-occurrence maps might not relate to any underlying division in the field, but rather to the fact that different types of studies use their own vocabulary. As such, any given author might publish both correlational and experimental studies, but use distinct terms, at least in the title, to describe these studies.

Discussion and Conclusion

This study aimed to critically evaluate the received view about the development of 20th-century American psychology. Using advanced text analysis tools, we quantitatively analyzed the metadata of 119,278 articles published in American psychology journals between 1920 and 1970. We extracted and analyzed the most-cited authors and the most

frequently occurring title words in each of the five consecutive decades. Furthermore, we generated cocitation and co-occurrence networks to visually map the similarity relations between psychologists and title words as well as to analyze the development of American psychology over time.

On the basis of these analyses, we submit that there are good reasons to suppose that the standard view needs reappraising: (a) In none of the decades that we analyzed did authors who are typically classified as behaviorists belong to the most prominent (combination of) clusters. Even on a very liberal definition of behaviorism—combining the citations of all the clusters that contain authors who are commonly classified as behaviorists (clusters which often also contained Gestalt psychologists, functional psychologists, developmental psychologists, and even cognitive psychologists)—the proportion of behaviorists was never larger than a third of the most-cited authors in a particular decade. Moreover, in all decades analyzed, measurement of personality and mental abilities seems to have been the main preoccupation of the field. (b) Our cocitation analyses show that the influence of behaviorism did not decline dramatically after the Second World War either (at least until 1970). Not only did the absolute number of citations to the most important behaviorists quadruple between 1940 and 1970, the share of citations to behaviorist articles also remained stable between 1940 and 1970 (approximately 28% of the field). (c) Finally, our analyses also challenge certain presuppositions about the development of behaviorism itself, suggesting, for example, that Skinner's work was less influential than the received view suggests.

Still, we would like to be cautious in our conclusions for both methodological and historiographical reasons. For one thing, there are several methodological limitations to the present study. First, we only analyzed two types of data: citation data and title words. There are many alternative data sources that might be useful to analyze the structure and the development of American psychology. We would be curious to see, for example, to what extent American graduate students were influenced by behaviorist approaches to psychology (e.g., by analyzing dissertations published between 1920 and 1970), to what extent behaviorism can be said to have been ideologically dominant (e.g., by studying textbook introductions to psychology between 1920 and 1970), and to what extent behaviorism played a role in American culture (e.g., by analyzing discussions about psychology outside academia).

A second methodological reason to be cautious is that our analysis is based exclusively on WoS data. Although our analysis of the coverage of WoS shows that its overall coverage is rather impressive, especially for major psychology journals, WoS only provides citation data for scientific articles. Because research monographs also played an important role in the development of American psychology, especially before the Second World War, our analyses do not take into account all academic output produced by American psychologists between 1920 and 1970. We do not have reason to presume that including data from scientific monographs would significantly affect our results, however, as our dataset does include citations to research monographs.²⁶ The only data that the present study does not analyze are the citations from research monographs. We do not have reason to believe that cocitation patterns in research monographs will paint a significantly different picture of the development of American psychology.

Second, our dataset only allowed us to construct cocitation networks on the basis of the first author of each publication. Taking coauthorship into account would have three interrelated consequences: (a) authors with a relatively large share of nonfirst authored publications would become more prominent in the networks, (b) the prominence of authors with relatively few nonfirst authored publications would be attenuated, and (c) the

²⁶ Indeed, research monographs such as Freud's (1913) *The Interpretation of Dreams* and Thorndike's *Teacher's Word Book* belong to the most-cited documents in our dataset.

distances between authors would change in ways that are difficult to predict. These are probably minor concerns, however, as single-authored articles were the norm for each of the decades under consideration.²⁷

Third, our analysis is based on a dataset comprising all English-language psychology articles published between 1920 and 1970, whereas our primary aim is to draw conclusions about the structure and development of American psychology. It is likely that our analyses somewhat misrepresent the structure of American psychology if there were significant differences between American psychology and psychology in non-American-English-speaking countries. It should be noted, however, that out of the top 25 journals with the most articles in our dataset (which together deliver 51% of the 119,278 articles in our catalogue), 24 are American journals.²⁸

Most importantly, however, we would like to be cautious in drawing any major conclusions about the rise and fall of behaviorism for historiographical reasons. Rather than claiming that our analyses refute the received view that behaviorism was the dominant approach in mid-20th-century psychology, we would prefer to use the results of our analyses to offer a broader perspective on the development of American psychology. For one thing, “dominance” is a notoriously vague concept. Indeed, the two main measures of influence examined in the present study—citations and title terms—paint subtly different pictures about the development of American psychology. Whereas our cocitation networks suggest that authors who are typically identified as behaviorists were less influential than the received view suggests, our term co-occurrence analyses reveal that behaviorist terms gradually became more prominent during the decades, thereby suggesting that behaviorist research might have played an important role in setting the direction of at least a part of psychological inquiry. And these are just two measures of “influence.” We can only imagine what still other (conceptual, sociological, educational, therapeutic, societal, or political) measures would show us about the way in which behaviorist research influenced the development of American psychology, American academia, and maybe even American society. Although Skinner’s influence appears to have been limited if one solely focuses on citations, for example, it is well possible that his work (a) influenced the conceptual development of American psychology (e.g., by developing the concept of operant conditioning), (b) contributed to the technological development of American psychology (e.g., by inventing the Skinner box), and (c) impacted American society because of his frequent media interviews, his contributions to debates about the ideal society (e.g., *Walden Two*), and his technical inventions (e.g., the air crib and the teaching machine; [Rutherford, 2003](#)). The story about the development of American psychology is a complex one and cannot be reduced to a single question about “dominance.”

Second, the term *behaviorism* itself is not a rigid, one-dimensional concept that can be defined using a fixed set of strictly delineated criteria. If anything, behaviorism is a broad collection of loosely associated psychologists and small research groups who share similar philosophical and empirical assumptions—assumptions, moreover, that changed significantly over time. For historians, it makes more sense to speak about similarities between authors, texts, and concepts, than to try to classify them using strict ahistorical categories.

²⁷ The percentages of publications by one author in the WoS dataset were 86%, 79%, 76%, 66%, and 55% for each of the consecutive decades. The co-citation networks were based on authors cited in these publications, but the numbers above provide a good estimate. If anything, because the co-citation analyses always “look back,” the percentage of single-authored articles in each of the networks is probably even higher.

²⁸ See [Figure 1](#). The exception is *Journal of Mental Science*, which is British. We decided not to exclude articles from British journals because we have insufficient information to determine the origin of all the 246 journals in our dataset. Furthermore, it should be noted that American psychologists also often published in British psychology journals (and vice versa), such that selection at the journal level will never deliver a clean-cut distinction between American and non-American Anglophone psychology.

Indeed, the present study shows that it is perfectly possible to study the development of behaviorist research without relying on a fixed, ahistorical definition of behaviorism. In fact, our study shows that relying on similarities between authors and concepts helps us to draw richer and subtler conclusions about the development of American psychology.

Finally, and most importantly, a too narrow focus on the rise and fall of behaviorism diverts the attention away from a broader perspective on the development of American psychology. Not only do our analyses show that American psychology was more diverse than the received view suggests, they also show that the story about the development of American psychology is mostly a story about a rapidly expanding field of study (growing from about 6,000 publications in the 1920s to more than 60,000 publications in the 1960s)—a story about the rise of a new and increasingly important discipline with ever broadening theoretical and applied concerns, characterized by a diversity of approaches. Our quantitative analyses, in sum, not only help us to reappraise the standard view about the development of behaviorism, they also show us that historians need to tell both a more diverse and a subtler story about the development of American psychology.

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